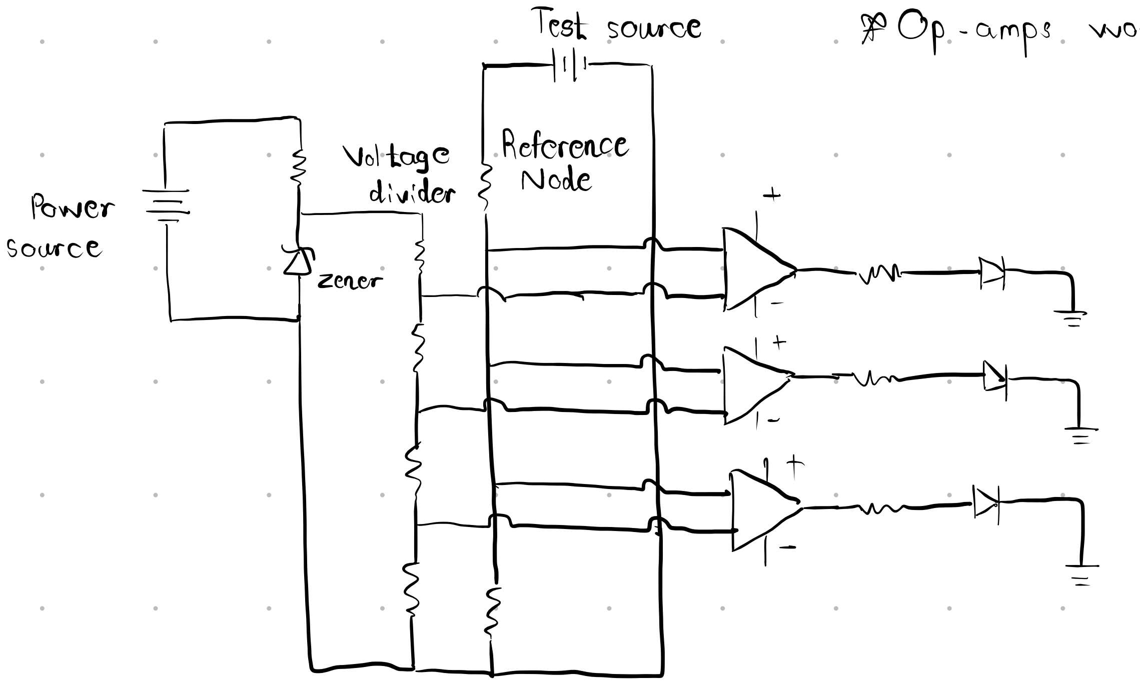
12-V Battery Level Indicator

Calculation



* Op-amps work on Dual-supply

Comparator rule

$V_+ > V_- \Rightarrow V_{out} = \text{high}$

$V_+ < V_- \Rightarrow V_{out} = \text{low}$

V_+ = Non-inverting $\Rightarrow$ Scaled battery voltage

V_- = inverting $\Rightarrow$ Reference ladder voltage

Why Zener ?

* Normal diode blocks reverse current.
* Zener is specially designed for reverse breakdown.
* When reverse voltage reaches breakdown voltage
  1. It conducts
  2. clamps voltage approximately constant

Zener Calculation

Supply = 12V

Zener = 5.1 V

Desired current ≈ 5 mA

$$R = \frac{V_{supply} - V_z}{I}$$

$$R = \frac{12 - 5.1}{0.005}$$

$$R = 1380\ \Omega$$

So, $R_2 = 1.3\ k\Omega$ (For assembly)

Why desired current ⇒ 5 mA ?

* The current zener needs depends on datasheet values.

Voltage ladder reference ⇒ 5100 Ω , V = 5.1V

$$I = \frac{V}{R} = \frac{5.1}{5100} = 0.001\ A = 1\ mA\ ✓$$

Zener gets ⇒ 4 mA ✓ okay

Battery scaling (Test source)

Reference is around only 5.1V (because of zener)

12V source needs to be scaled down.

Resistor choice, $R_1 = 1.5\ k\Omega$

$R_2 = 1\ k\Omega$

$$V = 12 \times \frac{R_2}{R_1 + R_2}$$

$$= 12 \times \frac{1}{1.5 + 1}$$

$$V = 4.8\ V$$

$$V = 11 \times \frac{R_2}{R_1 + R_2}$$

$$= 11 \times \frac{1}{1.5 + 1}$$

$$= 4.4\ V$$

12V	Scaled
1 V down	0.4 V down

So, resistor choice okay ⇒ 1.5k Ω and 1k Ω

Reference ladder divider

$$R = \frac{V}{I}$$

$$= \frac{5.1}{0.001}$$

$$R = 5100 \approx 5\,k\Omega$$

Each voltage step (scaled) from test source is 0.4 V.

LED red ⇒ less than 4 V (10V)

LED red + yellow ⇒ less than 4.4 V (11V)

LED red + yellow + green ⇒ less than 4.8V (12V)

$$5.1V - 0.3V = 4.8V$$

$$R = \frac{V}{I} = \frac{0.3}{0.001}$$

$$= 300\,\Omega \approx \text{use } 330\,\Omega$$

$$V = IR = 0.001 \times 330$$

$$= 0.33\,V \Leftarrow \text{voltage drop}$$

After R_1 ⇒ $V = 5.1 - 0.33$

$$= 4.77 \approx 4.8 \text{ (okay)}$$

$$R_2 = \frac{V}{I} = \frac{0.4}{0.001} = 400 \approx 390\,\Omega$$

$$R_3 = 390\,\Omega$$

$$R_4 = 5100 - (330 + 390 + 390)$$

$$= 5100 - (1110)$$

$$= 3990 \approx 3.9\,k\Omega$$

Resistor choice ⇒ 330 → 390 → 390 → 3900

Ladder current ⇒ 1 mA ?

It should be larger than op-amp input current.

LM741 ≈ around 80 nA.

1 mA >> 80 nA okay ✓

Why assembly values differ ⇒ 7, 8, 9 V

Diode we got around 4 V or more (Not sure)

Reference ladder ⇒ 4.2V − 0.3 ⇒ 3.9V
3.9V − 0.4 ⇒ 3.5V

Scaled ⇒ 10V ⇒ 4 V

∴ 9V ⇒ 3.6V

9.5V ⇒ 3.8V

9.75V ⇒ 3.9V

Threshold for Green become around 9.75 − 10

Threshold for yellow became around 8.75 which is 3.5 scaled.

Threshold for red became around 7.75.

In practice our leds light up at around 7, 8, 9.

So, the circuit concept is correct.

Note: Values can differ according to hardware.